

AccessMatrix Product Upgrade Guide - 6.0.1-GA (AI)

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1. Preface - AccessMatrix Product Upgrade Guide

This document contains guidelines for upgrading AccessMatrix.

Intended Reader

It is intended as a reference for system administrators and i-Sprint's Professional Service personnel when upgrading AccessMatrix.

Document Scope

This document highlights important considerations and the necessary steps for upgrading AccessMatrix.

For in-depth details and instructions on deploying the AccessMatrix system, refer to the [AccessMatrix Common Deployment Guide](#).

For feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- [AccessMatrix Release Notes](#)
- [AccessMatrix Common Deployment Guide](#)

2. Pre-Upgrade Requirements

This section contains the requirements needed for upgrading AccessMatrix.

Pre-Upgrade Checklist

Before upgrading AccessMatrix, it is recommended to review the tasks listed in the checklist below. It contains important considerations and reminders to ensure a successful upgrade.

Tasks	Reference
Review AccessMatrix Release Notes	Review AccessMatrix Release Notes
Review AccessMatrix Supported Platforms	Review AccessMatrix Supported Platforms
Verify AccessMatrix License	Verify AccessMatrix License
Consider SSL Setup	Consider SSL Setup
Backup AccessMatrix Installation and Database	Create Complete Backup

Review AccessMatrix Release Notes

Release notes are made available for each major AccessMatrix release. Ensure that you have reviewed the [AccessMatrix Release Notes](#) to familiarize with the changes introduced in the version you will be upgrading to. They contain important information, such as:

- New features
- Changes to existing features
- Vulnerability and major defect fixes
- Deprecated features

Review AccessMatrix Supported Platforms

The supported platforms document is made available for each major AccessMatrix release. Ensure that your deployment setup is compatible with the platforms listed in [AccessMatrix Supported Platforms](#) for the version you will be upgrading to.

i	Note: This includes ensuring that your Java environment is compatible.
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Verify AccessMatrix License

Ensure that your AccessMatrix license is still valid for the upgrade.

Consider SSL Setup

If a two-way SSL setup is used between the application and AccessMatrix servers, ensure that the application server supports TLS at a minimum. In addition, ensure that your SSL certificates are still valid.

Create Complete Backup

Before upgrading AccessMatrix, you must back up your existing AccessMatrix installation. A general guide is outlined below:

Restrict Access to AccessMatrix Server

To ensure that the data is unchanged during the upgrade process, you must temporarily restrict access to AccessMatrix. This prevents user actions from updating the data in the midst of an upgrade.

Backup Existing AccessMatrix Database

It is recommended to back up the existing AccessMatrix database before starting the upgrade process. This is important for failover operations, allowing for the backup database to be activated when issues occur during or after an upgrade.

Backup Installation Directory

Locate the AccessMatrix installation directory and create a backup in a separate directory.

Backup External Directories

If there are files or directories located outside the AccessMatrix installation directory, ensure that you back them up separately.

3. Upgrading AM5 to AM6

The AM6 series is designed to be upgraded from the AM5 series without requiring major configurations to be made to a new AccessMatrix server. As such, the upgrading process will be similar to upgrading within the same AM5 series. However, there are some important considerations to be made before doing so. This section highlights these considerations, and the required steps to ensure a successful AM6 series upgrade.

AM6 Series Upgrade Checklist

Before attempting the upgrade from AM5 series to AM6 series, it is recommended to review the tasks listed in the checklist below. It contains important considerations and reminders to ensure a successful upgrade.

Task	Reference
Review Upgrade Requirements	Pre-Upgrade Requirements
Consider Web Server	Web Server Considerations
Review Java Version	Review Java Version
Review AM5 Plugins and Web Applications	Review AM5 Plugins and Web Applications
Complete Upgrading Process	Upgrading AccessMatrix

Web Server Considerations

The AM6 series is designed to run on JBoss Web Server 6 / Tomcat 10.1, while AM5 series runs on JBoss Web Server 5 / Tomcat 9.0. AccessMatrix includes a bundled Tomcat 10.1 server that has been similarly hardened to previous versions.

- If you are planning to use the bundled Tomcat server, no additional changes are required.
 - If you are not planning to use the bundled Tomcat server, you must provision a new JBoss Web Server 6 / Tomcat 10.1 Server before attempting to upgrade from AM5 series to AM6 series.
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Review Java Version

AM6 series has been certified with Java 17 and Java 21. If you are using a version of Java that is lower than Java 17, you must upgrade your Java environment before attempting to upgrade from AM5 series to AM6 series.

Review AM5 Plugins and Web Applications

AM6 series is designed to run on Tomcat 10.1, which fully adopts the **Jakarta.*** namespace and removes all references to the **javax.*** namespace. This means that existing AM5 series plugins or java customization codes that are dependent on the **javax.*** namespace must be updated to be compatible with the AM6 series.

A Tomcat migration tool has been provided to aid applications in conforming to the new namespace requirements. It involves updating Java codes and application configurations to replace references to the **javax.*** namespace with **jakarta.*** namespaces. This automates changes for classes, libraries and deployment descriptors for convenience.

i **Note:** The migration tool is not available for third-party libraries and custom annotations extended with **javax.*** namespaces. These will either require changes to the namespace manually, or Jakarta-compatible replacements to be used.

Prerequisites for Tomcat Migration Tool

The following prerequisites are necessary to use the Tomcat Migration Tool:

- Java must be installed and properly configured in the system environment.
- Apache Tomcat Migration Tool must be downloaded and installed.

i **Note:** Refer to the official website for the [Apache Tomcat Migration Tool](#). To install, extract the downloaded ZIP file to a directory.

- Application files and plugins to be migrated must have a backup available.

Namespace Migration

To run the Apache Tomcat Migration Tool, perform the following:

1. Open the migration tool to display the command-line interface.
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2. Navigate to the directory containing the migration tool.
 3. Run the command: `java -jar jakartaee-migration-<version>.jar <source> <destination>`

**Note:**

- <source> refers to the file path of the WAR file, EAR file, or directory of the source code to be converted.
- <destination> refers to the file path where the converted file or directory should be saved.

4. Use WinMerge to compare between the original and processed files.
5. Review the updated files to ensure that all references have been migrated:
 - **Java Code/ JSP file/ Annotation:** Ensure all **javax.*** imports are replaced with **jakarta.***.
 - **Configuration files:** Ensure all XML descriptors and configuration files are updated to conform to Servlet 6.0 of Jakarta EE 10. For example:

XML

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"

         xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
         version="6.0"
         metadata-complete="true">
```

- **Libraries:** Ensure that libraries dependent on **javax.*** namespaces are replaced with Jakarta-compatible versions.



Note: As a convention, the names of migrated library **.jar** files should be appended with "-jakarta-EE". If you have changed the library files, ensure that you have an updated copy of the license.

6. Deploy the migrated application to a Jakarta EE 10-compatible application server (e.g. Tomcat 10.1.x, JBoss Web Server 6).
7. Test and ensure that the migrated application does not face runtime or compatibility issues.

Troubleshooting Common Issues

If the application does not work as expected after the migration, perform the following:

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- Check for manually written references to **javax.*** in code, configurations, and libraries.
 - Replace third-party dependencies that rely on **javax.*** with Jakarta-compatible versions.

4. Upgrading AccessMatrix

This section contains instructions on how to upgrade the AccessMatrix system.

i Note: Ensure that you have reviewed the [Pre-Upgrade Requirements](#) before attempting to do so.

Update Database Schema

Each AccessMatrix release may include updates to the database schema.

To update the existing AccessMatrix database schema that has been set up in previous AccessMatrix Installations, the database administrator will need to execute the relevant database schema update scripts provided.

! Important Note: Starting from 5.6.4-GA onwards, an additional database script to create a view of audit tables is to be executed. However, an error will occur if SQL Server Management Studio is used to run the script. To fix this problem, there is a need to update the script by adding "GO" statement before executing the create view script in SQL Server Management Studio (this step is not needed if you use other database clients like Squirrel SQL Client). For example:

PL/SQL

```
GO
createview am_vw_audit_b_a asselect*from am_audit_b
unionallselect*from am_audit_a;
```

To update the database schema, perform the following:

1. Execute the query script to check the schema version that is currently used for the database.

PL/SQL

```
Select va_value from am_vars where
va_name='SchemaVersion';
```

2. Open the update script corresponding to the database being used, e.g. **sqlserver-update.sql**.

i Note: The database script is located in the
<ACMATRIX_ROOT>/dbscripts/folder.

3. Edit the update script as necessary to include the relevant SQL statements for transitioning between the two schema versions.
 4. Save the changes.
 5. Execute the schema update script and commit.
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6. Execute the query script to validate that the schema version has been updated correctly. It must return a schema version corresponding with the upgraded Accessmatrix.

PL/SQL

```
Select va_value from am_vars where  
va_name='SchemaVersion';
```



Note: From 5.6.4-GA onwards, a script to perform audit data migration is to be executed separately after upgrading the database schema. On the same update script file (e.g. **sqlserver-update.sql**), uncomment and execute the following scripts to start moving the audit tables into the new tables:

PL/SQL

```
insert into am_audit_c ( au_server_id, au_id,  
au_reported_time, au_module_name, au_category,  
au_type, au_result, au_result_code, au_ext_result_code,  
au_message,  
au_tx_id, au_signature, au_actor_class, au_actor_uuid,  
au_actor_id,  
au_actor_user_store, au_actor_ses_id, au_actor_seg_uuid,  
au_actor_ip_addr, au_real_actor_id,  
au_real_actor_user_store, au_real_actor_ses_id,  
au_real_actor_seg_uuid, au_client_id, au_client_type,  
au_target_class, au_target_uuid, au_target_id,  
au_target_user_store, au_target_seg_uuid,  
au_custom_desc, au_attr1_char16, au_attr2_char16,  
au_attr3_char32, au_attr4_char32,  
au_attr5_char64, au_attr6_char64, au_attr7_char64,  
au_attr8_char128, au_flag1,  
au_flag2, au_flag3, au_flag4, au_attr1_num,  
au_attr2_num,  
au_attr3_num, au_attr4_num, au_time1, au_time2,  
au_time3, au_time4)  
select  
au_server_id, au_id, au_reported_time, au_module_name,  
au_category,  
au_type, au_result, au_result_code, au_ext_result_code,  
au_message,  
au_tx_id, au_signature, au_actor_class, au_actor_uuid,  
au_actor_id,  
au_actor_user_store, au_actor_ses_id, au_actor_seg_uuid,  
au_actor_ip_addr, au_real_actor_id,  
au_real_actor_user_store, au_real_actor_ses_id,  
au_real_actor_seg_uuid, au_client_id, au_client_type,  
au_target_class, au_target_uuid, au_target_id,  
au_target_user_store, au_target_seg_uuid,  
au_custom_desc, au_attr1_char16, au_attr2_char16,  
au_attr3_char32, au_attr4_char32,  
au_attr5_char64, au_attr6_char64, au_attr7_char64,  
au_attr8_char128, au_flag1,  
au_flag2, au_flag3, au_flag4, au_attr1_num,  
au_attr2_num,  
au_attr3_num, au_attr4_num, au_time1, au_time2,  
au_time3, au_time4
```

```
from am_audit;  
insert into am_audit_attr_c (aa_server_id, aa_id, aa_name,  
aa_part, aa_name_upper, aa_datatype, aa_value)  
select aa_server_id, aa_id, aa_name, aa_part,  
aa_name_upper, aa_datatype, aa_value  
from am_audit_attr;
```

Install Upgraded AccessMatrix Server

There are several ways to install the latest AccessMatrix server, depending on the web server that will be used to host the AccessMatrix instance. Refer to the following:

- For Tomcat Servers, refer to the [AccessMatrix Common Deployment Guide - AccessMatrix Installation](#).
- For Red Hat JBoss Web Servers, refer to the [AccessMatrix Common Deployment Guide - Deploy AccessMatrix on Red Hat JBoss Web Server](#).

Sync AccessMatrix Configurations

You will need to sync the AccessMatrix configurations if you have modified the configuration files of your existing AccessMatrix server installation. Ensure that you make the same modifications to your new AccessMatrix server installation.

The following table provides an example of the key configurations that must be synced:

File	Location in AxMx Tomcat Distributions	Location in AxMx WAR Distributions	Description
am5.xml	tomcat/conf/Catalina/localhost	Web server's JDBC Configuration	Refers to the database connection settings. This should be pointing to the existing database.
amsystem.properties	tomcat/webapps/am5/WEB-INF/classes	WEB-INF/classes	Refers to the am.server.id, data source, multicast settings, am.pluginidir (if applicable), among others.
web.xml	tomcat/webapps/am5/WEB-INF	WEB-INF	Refers to the services that are required to be

File	Location in AxMx Tomcat Distributions	Location in AxMx WAR Distributions	Description
			enabled if applicable, e.g. SOAP Services, SPNEGO, SAML, etc.
server.xml	tomcat/conf	Web server's Keystore Configuration	Refers to http/https connector related settings, i.e. type of connector, ports, server.keystore file path and passphrase, truststore.keystore file path and passphrase, sslprotocol, cipher strength.
amlog4j2.properties	tomcat/webapps/am5/WEB-INF/classes	WEB-INF/classes	Refers to the log level of am.log or whether there's specific class package level log level settings that's changed or added in.
am5-ehcache.xml	tomcat/webapps/am5/WEB-INF/classes/am5-ehcache.xml	WEB-INF/classes	Refers to the ehcache replication configuration on applicable components such as e2ee session cache, sms-otp, etc. (configure if applicable)

In addition to the files above, you should also consider performing the following configurations as part of the upgrade process:

Configuration	Instruction
Plugins	Copy any plugins directory and its contents that you had installed in the old AM5 folder to the new AM5 folder. For example, <i><ACMATRIX_ROOT>/tomcat/plugins</i>
Customizations	Copy any custom properties or .xml files from the old AM5 to the new AM5 classes directory. For example, <i><ACMATRIX_ROOT>/tomcat/webapps/am5/WEB-INF/classes</i>
JAR Files	Copy any custom .jar files deployed in the old AM5 to the new AM5 library directory. For example, <i><ACMATRIX_ROOT>/tomcat/webapps/am5/WEB-INF/lib</i>

Configuration	Instruction
Certificates	Copy any certificate containers (e.g. keystores / PEM files) from the old AM5 to the new AM5 certificate directory. For example, <code><ACMATRIX_ROOT> /tomcat/conf</code>
JDBC Driver	Copy the JDBC driver .jar files from the old AM5 to the new AM5 library folder. For example, <code><ACMATRIX_ROOT>/tomcat/lib</code>
Third-Party Libraries	Copy any third-party libraries from the old AM5 to the new AM5 library folders. For example, if you are using the OneSpan Authentication Suite Server SDK (formerly Authentication Server Framework or VACMAN Controller) with token libraries configured according to the UAS OneSpan Token Implementation Guide - Configure OneSpan Token Libraries , ensure that the various files follow the same filepaths in the new AccessMatrix installation.
JVM	Reconfigure the JVM arguments for heap size, etc, by making changes to the setenv.bat (Windows) or setenv.sh (Linux) files, if necessary. To confirm the changes, run the edited file via command line. Note: This is applicable only for AccessMatrix installations using the bundled Tomcat server.

Clean Up Existing AccessMatrix Installation

Once the new AccessMatrix version and has been set up, the previous AccessMatrix installation must be removed.

Perform the following steps:



Note: The following examples are illustrated for Windows environments using an AccessMatrix Tomcat Service named "AM_Server".

1. Launch **Registry Editor** and navigate to HKEY_LOCAL_MACHINE > SOFTWARE > WOW6432Node > Apache Software Foundation > Procrun 2.0 > AM_Server
2. Record the registry paths and keys of the existing Accessmatrix Tomcat Service.
3. Launch **Services** and ensure that the existing AccessMatrix server has stopped.
4. Record the **Startup Type** and **Log On As** columns for the existing Accessmatrix server.
5. Launch **Command Prompt** as an administrator.
6. Navigate to the directory `<ACMATRIX_ROOT>/tomcat/bin.`



Note: This file path contains the executable of the existing service.

7. Remove the existing AccessMatrix Tomcat Service by using the command: `service.bat remove AM_Server`

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8. In **Services**, ensure that the existing AccessMatrix Tomcat Service is no longer available.
 9. In **Registry Editor**, navigate to HKEY_LOCAL_MACHINE > SOFTWARE > WOW6432Node > Apache Software Foundation > Procrun 2.0 > AM_Server
 10. Ensure that the registry key of the existing AccessMatrix Tomcat Service is no longer available.
 11. Backup the existing AccessMatrix installation path in <ACMATRIX_ROOT>.
 12. Remove the existing AccessMatrix installation path in <ACMATRIX_ROOT>.

Post-Upgrade Validation

Once AccessMatrix has been upgraded, it is essential to complete post-upgrade checks to ensure that the upgrade is successful.

Start Up AccessMatrix Server

Confirm that the upgraded AccessMatrix server and all relevant plugins and modules can be started by performing the following:

1. Start the AccessMatrix service.
2. Navigate to the directory <ACMATRIX_ROOT>/tomcat/logs and open the **am.log** file.
3. Ensure that the **am.log** file contains specific keywords.

 **Note:** For more details, refer to [Log File Keywords](#).

Log File Keywords

The log file can be used to verify that the AccessMatrix service and any other deployment specific plugins have been started correctly.

 **Note:** The following list is not meant to be exhaustive. If you are unsure of the relevant keywords, contact i-Sprint's Professional Services.

Item	Keywords
AccessMatrix Version	AccessMatrix(TM) version "6.0.0.0029-GA-E01"
Server Started	Server Context Started. Version 6.0.0.0029-GA-E01

Validate AccessMatrix Server and Deployment Specific Modules

Once the AccessMatrix server has been started, ensure that essential configurations can be made using the admin console. The required modules should be functioning as expected.

Perform the following:



Note: While the validation process can be specific to each deployment, the following serves as a general guideline.

1. Start the AccessMatrix server.
2. Review the **am.log** files for error messages.



Note: If there are errors that cannot be resolved, contact i-Sprint's Professional Services and attach the relevant log files.

3. Access the Admin Console login URL (e.g. <http://localhost:8080/am5>) and ensure that the page displays the correct version number.
 4. Ensure that you can log in using an existing administrator account.
 5. Configure and validate that the required modules are working correctly.
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5. Upgrading with Efixes

Efixes are minor patches that contain defect fixes or minor enhancements. They can be applied to existing AccessMatrix versions without major changes required.

A general guide for applying the efix patch is outlined below:

1. Backup the existing AccessMatrix installation.



Note: For more details, refer to [Create Complete Backup](#).

2. Extract the efix package provided by i-Sprint's Professional Services. It contains a readme and other relevant files for the upgrade.
3. Follow the instructions provided in the readme.
4. Test and validate the upgraded AccessMatrix server before moving it to production.

6. Known Issues and Troubleshooting

If there are issues with the upgrading process, troubleshoot the issue using the guidelines provided in this topic. To rollback an AccessMatrix upgrade, refer to [Rolling Back AccessMatrix Upgrade](#).

Failure to Start AccessMatrix with Database Synonyms

When using a database update script as part of an upgrade to the AM6 series, the use of database synonyms in Oracle databases can potentially prevent the AccessMatrix server from starting up.

For more details on this issue and how to resolve it, refer to the [AccessMatrix Common Deployment Guide - Failure to Start AccessMatrix with Database Synonyms](#).

Breaking Changes in OneSpan Authentication Suite Server SDK Upgrade

Depending on the OneSpan Authentication Suite Server SDK (A3S, formerly known as VACMAN Controller or OneSpan Authentication Server Framework) version that is being upgraded, issues with backward compatibility can be introduced. Once A3S has been upgraded, token transactions that are made (e.g. OTP verification) will be stored in the AccessMatrix database using updated schemas. New token blob data that has been generated may not be recognized by prior versions of A3S.

As such, attempting to roll back the A3S upgrade once the updated token blob data is stored in the database can cause errors once the affected token undergoes a transaction. It can be identified by the following error:

Code
2025-05-05 15:15:58.123 [https-jsse-nio-1234-exec-10] ERROR - (GenericTokenService.java:319) error: 1118 Unsupported Blob; Unsupported Blob

To resolve this issue, the AccessMatrix database will need to be restored from a previous backup.



Note: Token blob data is stored in the am_tokenauthstate table.

Rolling Back AccessMatrix Upgrade

In the event where there is an issue with the AccessMatrix upgrade, the previous AccessMatrix version can be restored by performing the rollback procedures. A general guide for rolling back AccessMatrix upgrades is outlined below:



Note: Any changes that are made since the last backup will not be available after rolling back the upgrade.

1. Stop the upgraded AccessMatrix server and start the AccessMatrix server to be restored.
2. Restore the AccessMatrix database backup using a preferred database server tool.
3. Ensure that the correct AccessMatrix server has started.
4. Complete the steps in [Post-Upgrade Validation](#).
